Aravind Sudheer

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PROFILE

Robotics and mechatronics engineering graduate with professional ABB programming experience and project-based ML and computer vision work. Implemented content-based recommendations, pretrained vision integration and coursework in regression, clustering and image processing.

TECHNICAL SKILLS

Programming and tools: Python, NumPy, pandas, scikit-learn, OpenCV, MediaPipe, Streamlit

Methods: TF-IDF, cosine similarity, regression, clustering, facial geometry, head-pose estimation

Engineering: ABB robots, RAPID, RobotStudio, calibration, system troubleshooting

SELECTED PROJECTS

Content-Based Movie Recommendation System

Python, pandas, scikit-learn, Streamlit

- Implemented a combined-text TF-IDF baseline and a weighted model across genres, keywords, overview, director and cast.
- Added metadata validation and feature-score explanations; documented hand-selected weights and the absence of relevance evaluation.

Driver Monitoring System — Computer Vision Prototype

Python, OpenCV, MediaPipe, NumPy

- Combined pretrained facial landmarks, eye/mouth geometry and temporal rules to monitor sustained eye closure and head movement.
- Implemented solvePnP-based approximate head pose and neutral calibration, with synthetic geometry and monitoring-state tests.

Image Stitching — Coursework

Python, NumPy, OpenCV, matplotlib

- Implemented ORB feature matching and RANSAC homography estimation, with bilinear interpolation and alpha blending to combine image pairs.
- Generated stitched-image visualizations for supplied and additional image pairs.

PROFESSIONAL EXPERIENCE

Junior Robotics Engineer

Simpliforge Creations | Hyderabad, India | Dec 2025 – May 2026

- Programmed and simulated ABB six-axis robots using RAPID and RobotStudio for concrete 3D printing.
- Configured and calibrated workobjects, tool data and end-effectors before robot-program execution.
- Troubleshot robot-motion and program-execution issues during live printing; monitored PLC signals and process parameters.

Design & Application Engineer (Intern)

Automation Systems and Services | Kuala Lumpur, Malaysia | Nov 2023 – Feb 2024

- Supported industrial-system commissioning through PLC signal checks, electrical validation, WinCC layouts and documentation.

EDUCATION

Bachelor of Engineering (Honours) — Robotics and Mechatronics Engineering

Monash University | Graduated July 2025